CleantechHUB
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ENERGY COOP

CleantechHUB
Energy communities

Partner against energy poverty: datos públicos, cooperativas de O&M y ecosistema cleantech colombiano.

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Foto: Tierra Grata · rural communities, Colombia

CURRENT

CleantechHUB — overview

Climate innovation hub for Latin America para América Latina: conecta startups cleantech con corporativos, inversores, multilaterales y gobierno.

- **Since 2019** impulsando emprendimiento climático y cleantech (cleantechhub.net).
- **Countries cited on website:** Colombia, Perú, El Salvador, Guatemala, Costa Rica.
- **Platforms:** institutional website, wiki (BookStack), portafolio Nexus, decks y demos públicos.

Source: JSON-LD Organization, cleantechhub.net (oct 2026).

CURRENT

ClimateLaunchpad / CLP26

CleantechHUB es National Organizer de la competencia global de ideas verdes.

- **CLP26 campaign** en Colombia, Perú, El Salvador, Guatemala y Costa Rica.
- Call and acceleration de ideas de negocio verde → vía al portafolio Nexus.
- Public deck de campaña: **decks.cleantechhub.net/cth-clp26/**

Source: skills/clp26-brand/SKILL.md; CLAUDE.md (National Organizer).

CURRENT

Nexus — startup portfolio

Bilingual profiles (EN/ES) de alumni cleantech — energía, agua, economía circular.

- **nexus.cleantechhub.net** — fichas por startup (problema, solución, impacto, contacto).
- Fuente primaria de datos: wiki BookStack; badges ClimateLaunchpad por cohorte.
- Energy Coop destaca Sunco, SEOS, Soluna, Tierra Grata y TuWatt del ecosistema.

Source: skills/nexus-onepager/SKILL.md; brochure-v2.

CURRENT

Energy Coop

Energy cooperatives to close the gap de acceso y sostenibilidad O&M en Colombia.

- Social layer + contratos OPEX alineados CREG/SSPD — sin competir por fondos MW ni greenfield.
- Data lab + candidate territoriales (PASO en diálogo; PASO no ejecuta energía).
- Demo interactivo: **energiacoop.cleantechhub.net/demo**

Source: docs/energy-coop/brochure-v2/brochure-es.html; services/energy-coop/README.md.

CURRENT

REIN Hubs

Resilience · Entrepreneurship · Innovation · Nature — red de hubs físicos que activan, incuban y aceleran founders cleantech.

- CleantechHUB como socio de ejecución junto a PVBLIC Foundation.
- Programas en LATAM; vía ClimateLaunchpad como escalera de competición global.
- Plataformas de aprendizaje y wiki citadas en propuestas (sin cifras presupuestales en este deck).

Source: decks/rein-academic-city-ghana/index.html (contexto REIN); AGENTS.md (no montos REIN).

PRIOR

P4G / MubOn

Competitive line P4G (Partnership for Green Growth) — archivo de propuestas y packs MubOn en Drive CTH.

- Co-financiamiento estricto 50/50 en convocatorias P4G (regla grant desk).
- Integración con grafo de grants UNDP · P4G/MubOn en documentación interna.
- Sin cifras de adjudicación citadas aquí — solo trazabilidad de programa.

Source: skills/cth-grant/SKILL.md; docs/grant-graph/spec.md.

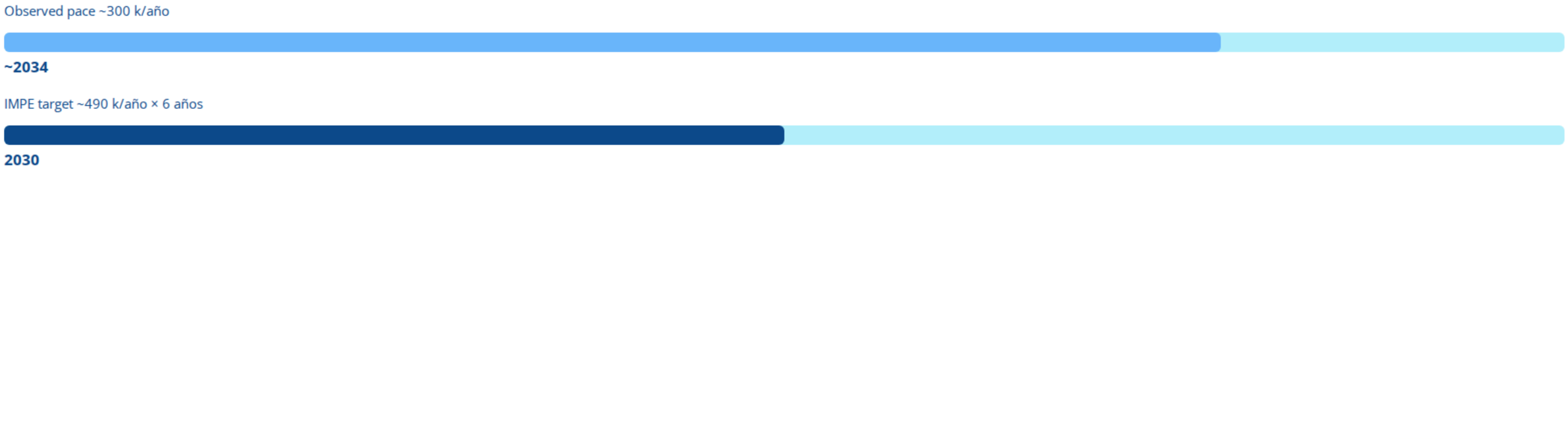
The gap and OPEX

15,4%
IMPE 2024 (~8,1 M pers.)

9,5%
Meta 2030 (~5,17 M; −2,9 M)

75,9% / 93,1%
ICEE rural vs nacional 2024

63,1% / 52,8%
IMPE La Guajira / Chocó



Contraloría
Sin remunerar operación y AOM de microredes y SFV.

Amazonas
~$30.000 M (regalías) en FV sin operar; baterías dañadas.

CREG
Miles de SFV rurales sin gestión → abandono.

CTH differentiator: hacemos *bankable* la capa social y OPEX donde hay ejecutor y CAPEX comprometido.

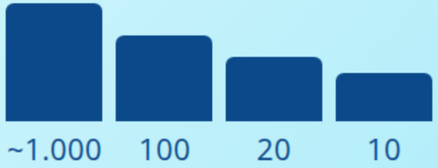
Source: IMPE 2025; UPME ICEE 2024; brochure-v2 notas al pie.

Cooperative + ecosystem

CTH proposal: O&M comunitario, ingresos locales y gobernanza cooperativa; KPIs dimensiones IMPE.

Sunco	**SEOS**	**Soluna**	**Tierra Grata**	**TuWatt**
898 kWp · DISPAC La Guajira[9]	+75 SOLvers · 18 dept.[10]	Prepago ZNI Caribe/Orinoquía	1.947 soluciones energía[11]	Energy communities

Data lab (illustrative): ~1.000 → 100 → 20 → 10 candidates · Corozal, Sucre candidate (IMPE 43,1%).



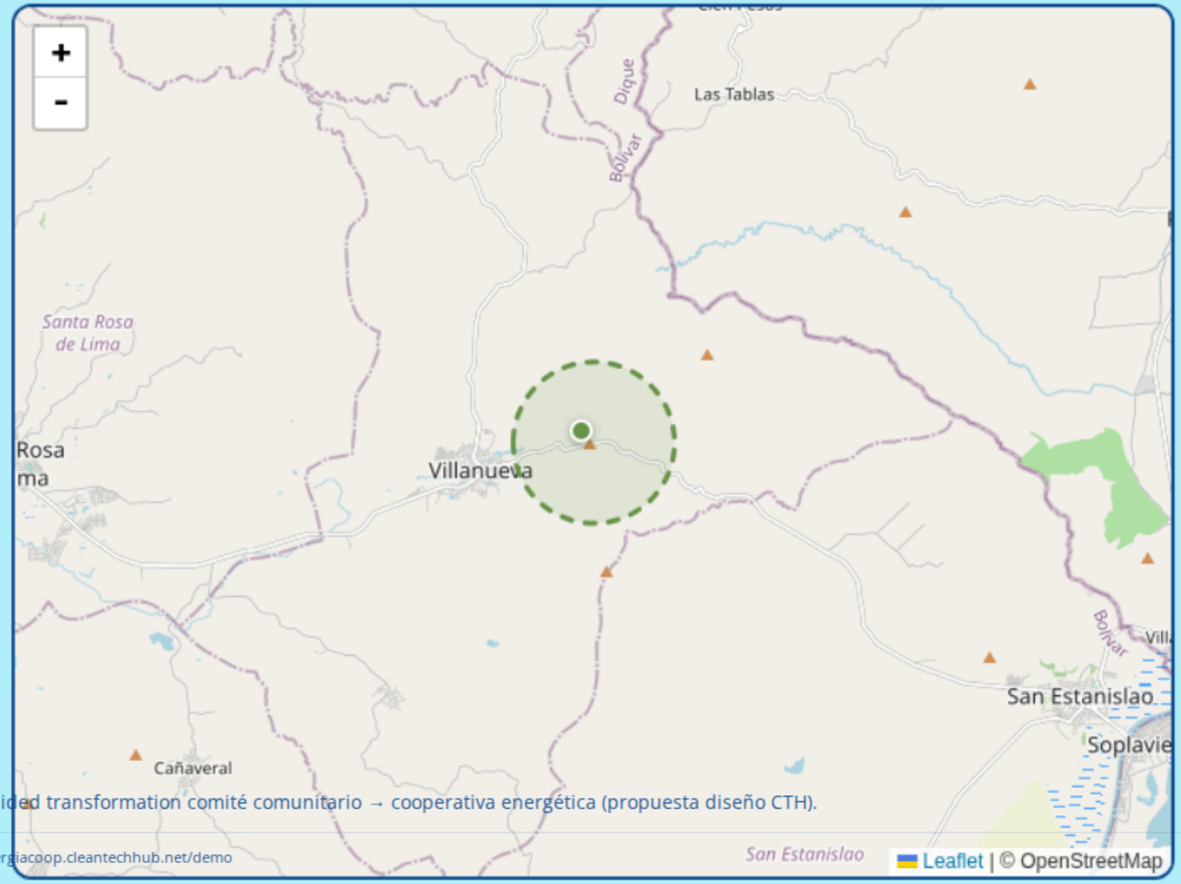
Source: brochure-v2; métricas startups según notas 9–11.

Paiva: from committee to cooperative

Community solar energy in vereda Paiva, Bolívar (~450 inhabitants), IE Chiricoco. NOT FENOGE Map location **approximate**.

National energy poverty (IMPE, index on persons): 15.4% · ~8.1 M people (2024)
Informe IMPE 2025 (PDF) · Nota IMPE, nov 2025

Rural electricity coverage 75.92% vs national 93.12% (UPME ICEE 2024)
Boletín ICEE 2024 (UPME) · UPME, noticia 2024



Guided transformation comité comunitario → cooperativa energética (propuesta diseño CTH).

PUBLIC SOURCE

Community solar project with local committee and IE Chiricoco (linked public sources).

Noticia Paiva (2025) · Guía del Gas

CTH DESIGN PROPOSAL

Guided transition: community committee → energy cooperative with ~30% participation in the community share class (CTH proposal, not a contractual figure).

Community committee
Grassroots organization

→

Energy cooperative
Legal personality

Transition shown

- General assembly — supreme social body
- Board of directors — oversight and strategy
- Operator — technical O&M of the asset
- Energy-sector ally — know-how (role per local agreement)
- Community share class — ~30% CTH proposal in capital structure

~450 inhabitants and territory: public sources (links). Map coordinates: approximate (no geocode in source).

La Guajira a escala

CleantechHUB Energy Coop

Energy Coop Walkthrough

ES EN

La Guajira at scale

Public projects in La Guajira (FENOGE and related processes). Pins without official coordinates: *approximate location*.
Verification: `docs/energy-coop/demo-cth-brand/SOURCES.md`.

IMPE La Guajira 63.1% (2024)
Informe IMPE 2025 (PDF) · PARTIAL — 1ST DEPT. SOURCE

ICEE La Guajira 54.28% (UPME 2024)
Boletín ICEE 2024 · UPME, noticia

Cooking with firewood/charcoal/waste: 56.6% of people in energy poverty (IMPE 2025)
Informe IMPE 2025 (PDF) · Resumen ejecutivo IMPE 2025

0	0	0
Projects on map	Verified kWp (sum)	Announced MW (utility-scale)

View La Guajira at scale



Data lab — funnel

Data Lab — prioritization funnel

How the Energy Coop pipeline narrows candidates before PASO. Public policy figures and documented cross-checks from `docs/energy-coop/`; wave counts (100 → 20 → 10) are EXAMPLE DATA illustrating the flow.

- **Public universe (reference)**
  National target of **500 energy communities 2024–2025**, 40% ZNI (FENOGE). In La Guajira, IPSE reports **148 EC** and **1,409 families** under Miichi Ka'i (Dec 2024).
  Sources: `energy-poverty-kpis-2026-11.md` · FENOGE · IPSE
- **IMPE / ICEE / IPEM cross-check**
  Desk layer: departmental IMPE incidence (e.g. La Guajira 63.1%, Chocó 52.8%), rural vs urban ICEE, household IPEM (different index). Does not replace field visits.
  Cross-check: `energy-poverty-kpis-2026-11.md` · IMPE 2025 PDF
- **Solar resource layer (screening)**
  Google Solar BASE where coverage exists (e.g. Corozal, Uribia); NASA POWER / Global Solar Atlas where not (Quibdó, Bahía Solano). Roof signal, not ground truth.
  `solar-api-coverage-2026-11.md` · `solar-api-integration-plan.md`

EXAMPLE DATA

100

Candidates after public cross-check + preliminary scoring

↓

20

PASO — candidates productivas

PASO — productive community candidates

Territories in dialogue for productive cooperative models. Locations and candidate status only.

CANDIDATE COMMUNITY

Corozal, Sucre

Municipal seat — reference point with Google Solar BASE coverage in map check (9.31778, -75.29583).
`solar-api-coverage-2026-11.md`

Productive communities in dialogue (territorial name only):

CANDIDATE COMMUNITY
Fonseca, La Guajira

CANDIDATE COMMUNITY
Puerto Asís / Guzmán, Putumayo

CANDIDATE COMMUNITY
La Macarena, Meta

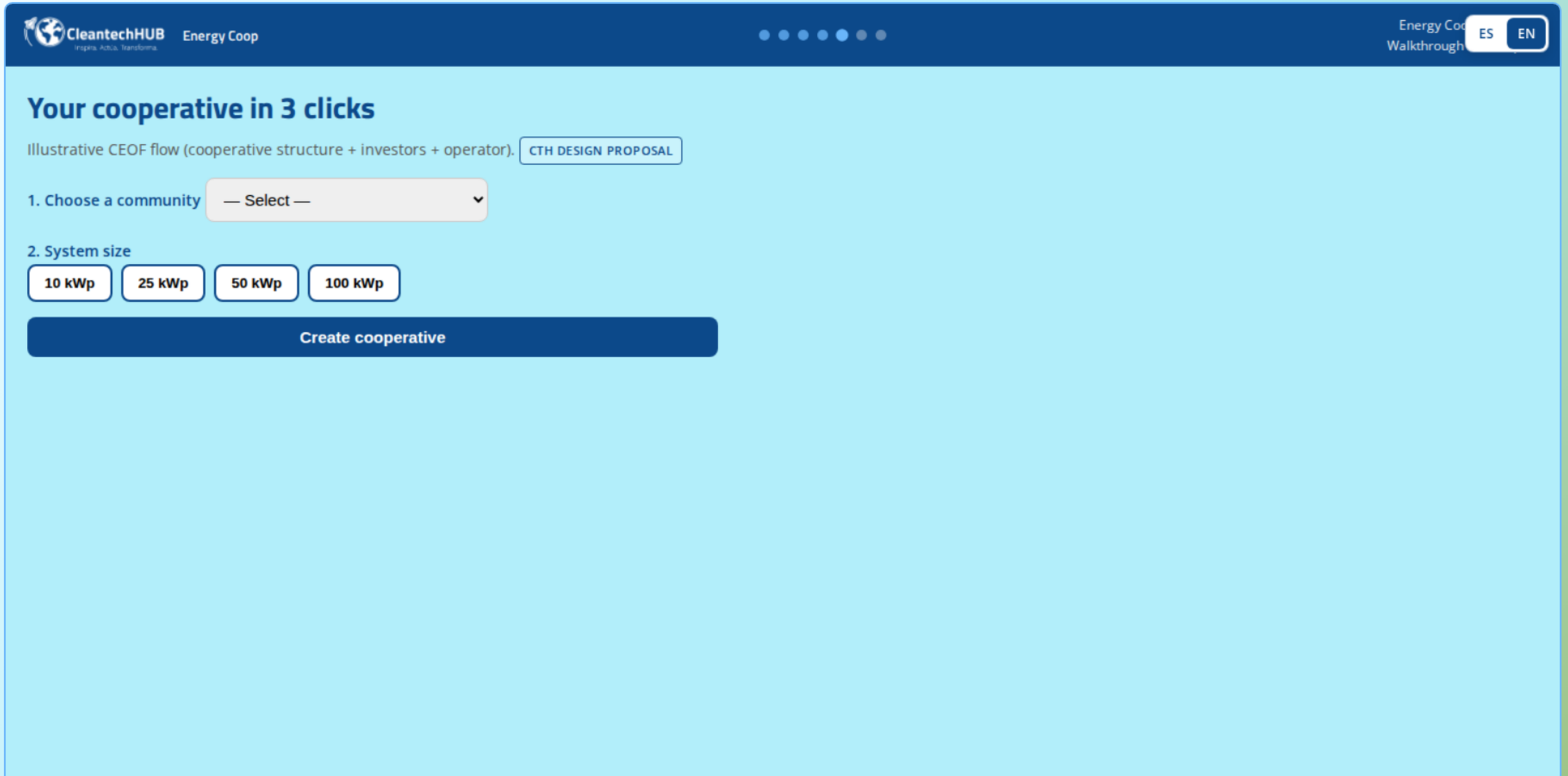
CANDIDATE COMMUNITY
Istmina, Chocó

CANDIDATE COMMUNITY
San José del Guaviare, Guaviare

CANDIDATE COMMUNITY
Miranda, Cauca

CANDIDATE COMMUNITY
Arauca, Arauca

Tu cooperativa en 3 clics



Meta 2030 + OPEX

2030 target + local talent

National IMPE scenario. Index on **persons**. Sources: `energy-poverty-kpis-2026-11.md`.

IMPE 2030 target: 15.4% (8.11 M people, 2024) → 9.5% (5.17 M) in 2030 · −2.9 M people

Informe IMPE 2025, p. 33 (PDF) · Nota IMPE, nov 2025

Pace gap (cited arithmetic)

People who must leave energy poverty ≈ **2.9 M** (8.11 M − 5.17 M; IMPE 2025 p. 33).

Observed pace ~300,000 people/year

`2,900,000 ÷ 300,000 ≈ 9.7 years from 2024`

~2034

IMPE 2025 p. 12 (~300 k/year) · IMPE 2025 report (PDF)

Target scenario pace ~490,000 people/year × 6 years

`490,000 × 6 = 2,940,000 ≈ 2.9 M`

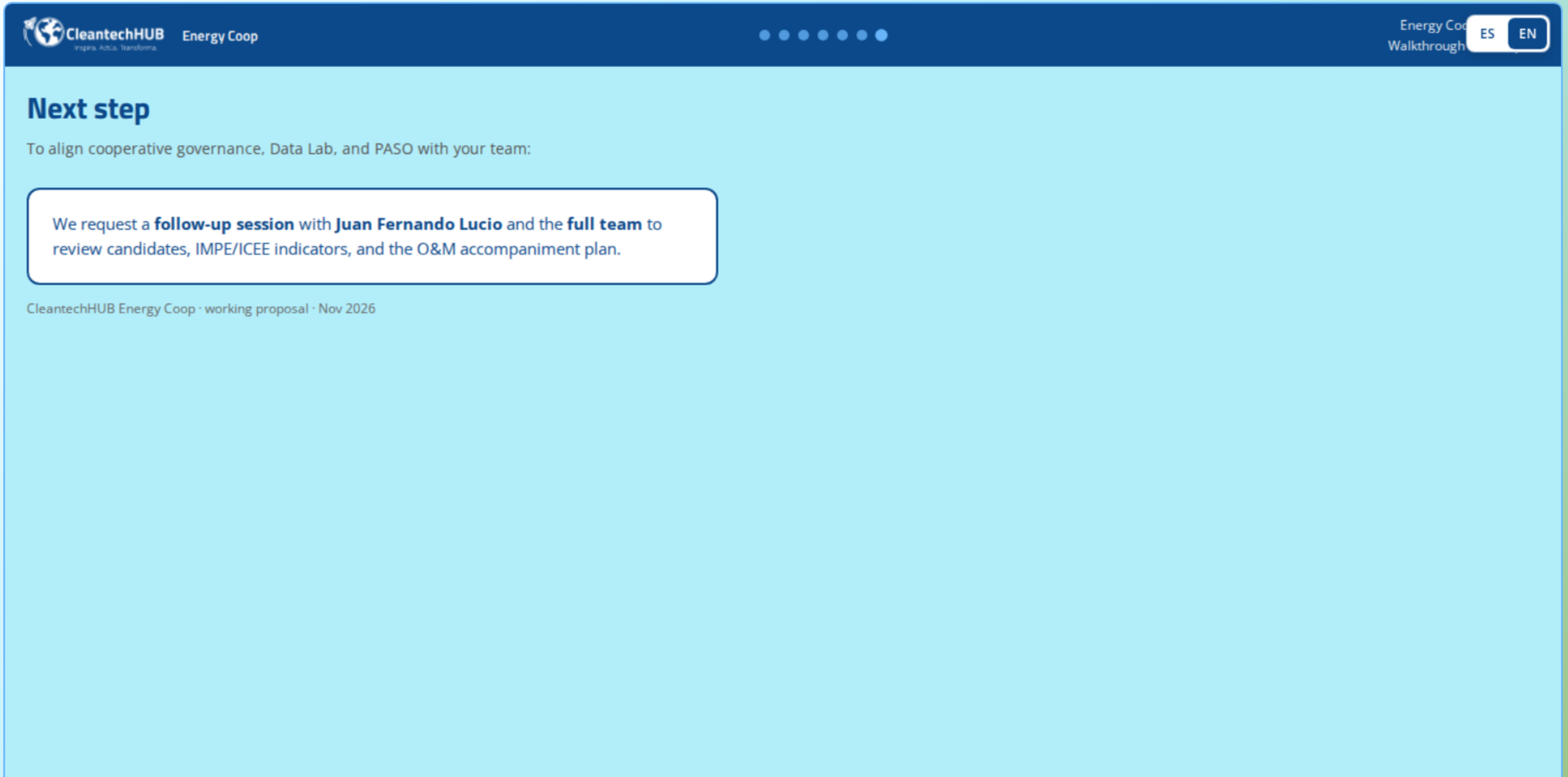
2030

IMPE 2025 p. 33 (≥1.0 p.p./year incidence)

Not a CAPEX problem, it is OPEX

- Contraloría: no methodologies to remunerate operation, microgrid administration and SFVI → maintenance and monitoring failures. El Colombiano
- Amazonas (~$30,000 million in royalties): abandoned photovoltaic project; damaged batteries — "critical state" due to sustainability. Seguimiento.co / Procuraduría
- CREG (GEN-Renovables): thousands of rural PV systems without management → "often abandoned"; missing regulation, certified installers and O&M. Circular 005-13 informe final · Concepto CREG 4022/2019 (AOM)

Proposed follow-up (demo UI) (UI demo)



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Live demo

docs/energy-coop/demo-cth-brand → energiacoop.cleantechhub.net/demo

← → · Espacio · Recorrido ~7 screens

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